



Royal University of Phnom Penh

Faculty of Engineering

Major: Information Technology Engineering

Kampu Property Rental Web application

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1. Introduction

Background

The rapid growth in population in urban areas around the world has been widely observed by the researchers. Phnom Penh, the capital of Cambodia is one of the fastest developing cities in the region. As urbanization continues to expand, one of the major challenges faced by the residents is the lack of accessibility to reliable housing information.

Students, workers, newcomers often encounter difficulties in finding a suitable accommodation due to the limited available information and lack of trust in listings on social media platforms. This situation leads to inefficiency, uncertainty, and potential risks when searching for rental properties in the city.

To solve this issue, this project aims to increase the overall user experiences for both parties, by implementing the web-based application, where it acts as a marketplace for both parties. The lessor can efficiently manage their rental records, advertising their rental properties, and increase their revenues . As for the user, they are allowed to browse the available properties based on their own preferences, and to contact the property owner .

Problem Statement

- High chance of being scammed through unverified lessors
- Traditional way of searching for properties leads to time wasting and energy
- Incomplete or unreliable property information
- Difficulty in finding rooms near a specific location

Objectives

The main aim of this project is to design and develop a **Web-based application for renting property** that helps improve the efficiency of the operations and to provide convenient renting process experiences for the end users.

The following are the mains objectives of our system:

- To provide all the essential property informations
- To provide a reliable and safe environment for the lessor and the lessee
- To support the students, workers, newcomer to find a suitable accommodation based on the budget
- Improving the user trust through the verification systems

Scope

This system allows both national and international users as well as the property owners to utilize the available functions through our web-based application based on their roles.

- + Available
 - User registration via Email or Phone number with ID/Passport verification
 - Rooms listing
 - Add, Update, View, Edit, Delete the listing properties
 - Find roommates
 - Admin Dashboard
- + Not implemented
 - Payment gateway via the web
 - Communication channel

Stakeholders

During our development phase, we will need a few stakeholders to involved to make our development progress goes smoothly, we have defined our important stakeholders as described below:

- **Customer:** End user who will be using the system for browsing and finding a suitable room.
- **Landlord:** One of our target audiences, we will be doing a UX to get the essential details about their needs
- **System admin:** Since this system needs the verification process, we will need an admin who will be monitoring the overall system activities
- **IT Maintainer:** A technical person, who will be responsible for the non-functionalities, to ensure a smooth business operation
- **Development team:** This team consists of a project manager, designers, developers, and testers.

Functionality

This web-based application provides the features based on the roles. The features will be described as follow:

- + Customer
 - Create an account via phone number or emails and required an national ID card or passport for foreigner
 - Login via registered method (phone or email) and passwords
 - Browse through the available properties, filter by prices, location, and type of renting property
 - Contact a landlord via an external link
 - Finding a roommate to share the living expenses
- + Property Owner
 - Listing their property for rent
 - Going through a verified process before the listed property reach audience
 - See the listing performance summary on a dashboard
 - Make a subscription to reach a wider audience
- + Admin
 - Admin Dashboard
 - Performing the verifications processes
 - CRUD operation on all the users inside of the system

Benefits

- Students who had just moved to continue their study can find a numerous of suitable properties based on their preferences
- Confidential information is secured
- We offer a solution related to living expenses, through one unique function called “Find mate”. It is a function which allows the user to find the living mates who they wish to share the property rental expenses with.
- Can conveniently scroll through the properties within just a device that connects to the Internet
- All properties information is displayed on the screen
- Can find a property location within their search area

Limitations

- We do not have the control over the fraud behaviors act by any party outside of application environment
- No control over the properties rules or renting contract information
- Lack of Mobile Application. This phase will be implemented later.
- No authority over the security matter, all the parties must be self-awareness

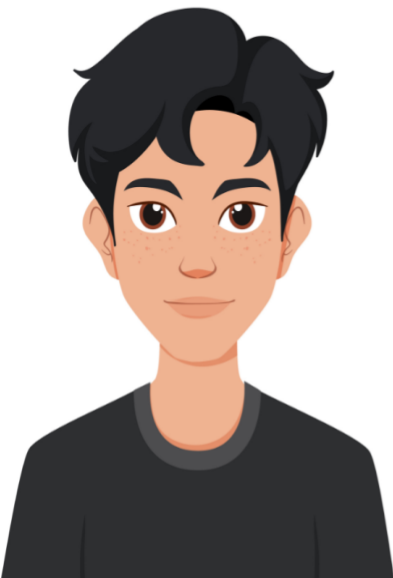
2. System Analysis

User Persona

SIDETH PANHA

PROFILE

Gender : Male
Age : 20
Education : Bachelor's degree
Occupation : Student
Address : Phnom Penh, Cambodia



BIOGRAPHY

Panha is a student who is currently pursuing his bachelor of English at Institute of Foreign Languages, RUPP. He comes from Kampong Cham to continue his higher education in the city.

GOALS

- Wants an efficient way to find a suitable accommodation
- Wants to find a suitable room with an affordable price
- Wants to know all the room details before moving in
- Can communicate in a safe environment without any worries

FRUSTRATIONS

- Having to ride motorcycles around to find a rental place
- There are many fraud lessor on the social media advertising
- Worried about getting scams because of the lack of information or rental rules

NEED

- Needs a digital way of finding a rental place
- Receiving all the essential rental rules before checking in the room

PERSONALITY

Introvert —●— Extrovert
Thinking —●— Feeling
Sensing —●— Intuition

TECHNOLOGY

Website
Social Media
Mobile App

Lessee

User Persona

Ms. Sreymom owns more than 20 rental rooms near university. She is currently advertising her property through many social media platforms. Having to answer the inquiries and updating rooms status daily is such a time consuming and disorganized.

Name: Srey Mom

Age and gender: 44 year old, Female

Job: Property Rental Owner

Location: Phnom Penh



Need

Goals:

- A platform to list and manage the properties more efficient
- The ability to update the room status instantly
- A function to list all the essential details
- To have a messaging portal with verified potential customers

GOALS & CHALLENGES

Frustration

- Too many repetitive messages from different platforms
- Customers asking the same questions such as price, location, and availability
- Difficulty in verifying serious renters
- Hard to keep listings upodated manually

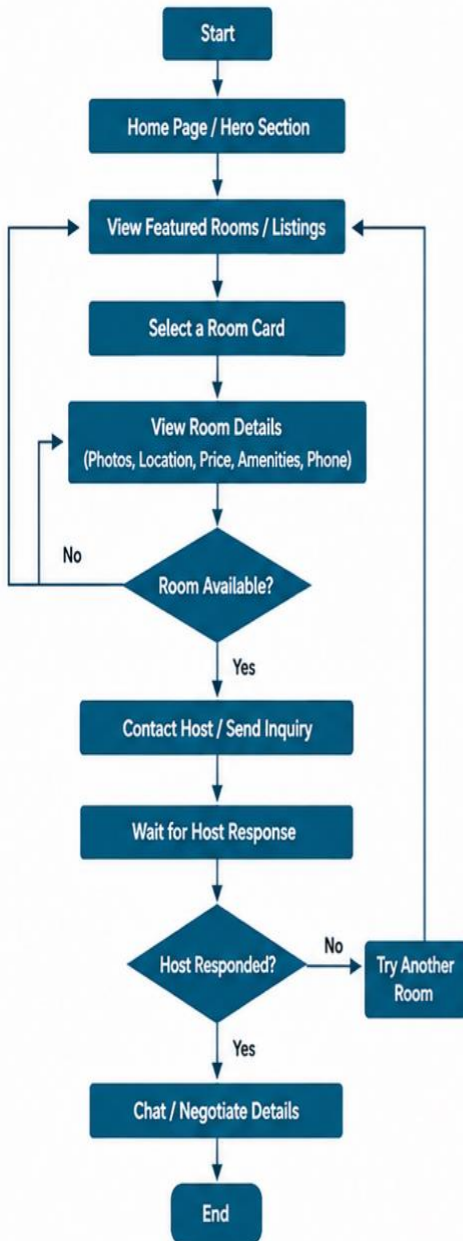
Goals:

- Find tenants quickly and consistently
- Promote property to a wider and targeted audience
- Managing the listings easily in one place
- Communicate efficiently with potential tenants

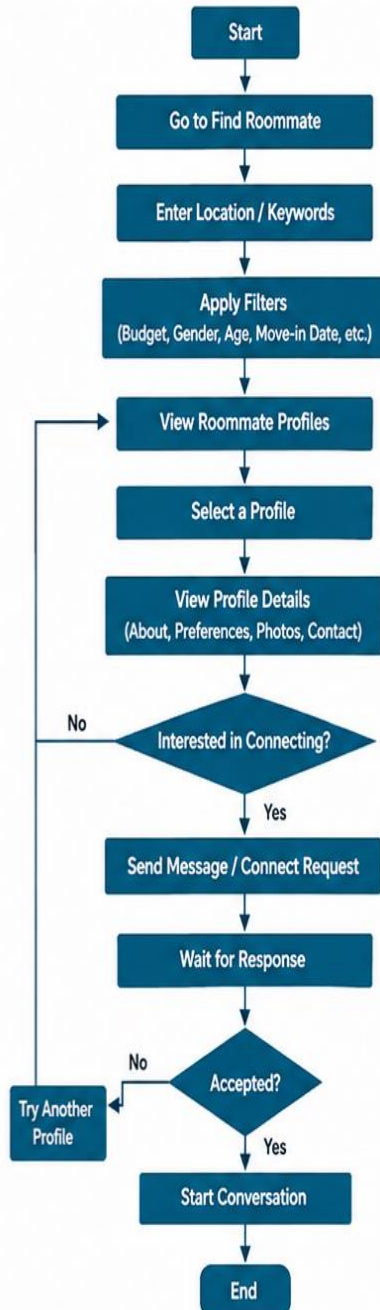
Lessor

Flowcharts

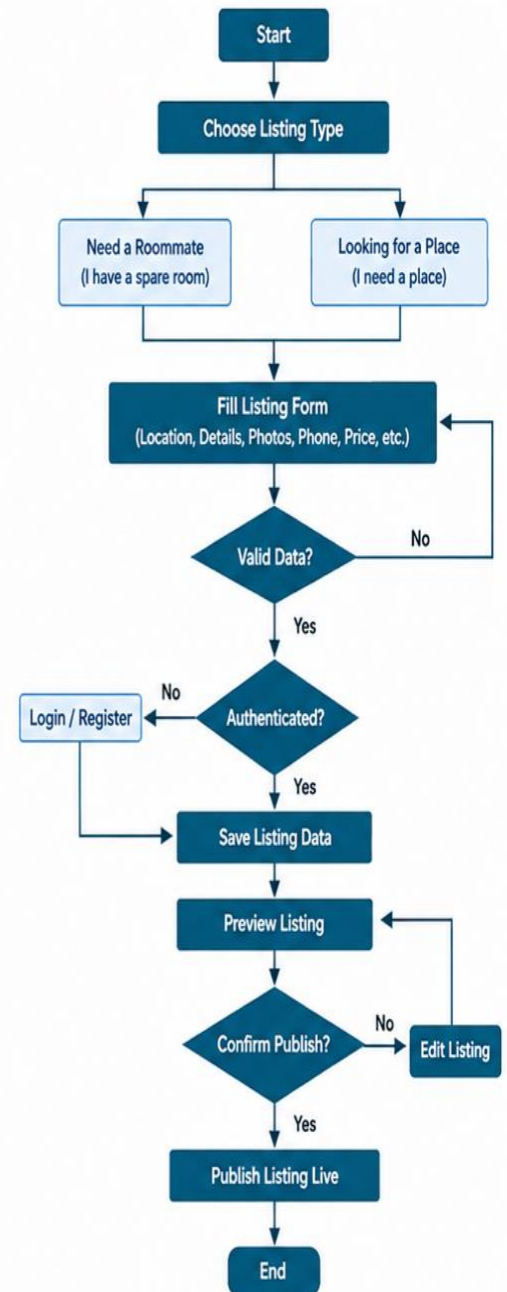
1. BROWSE ROOM FLOW

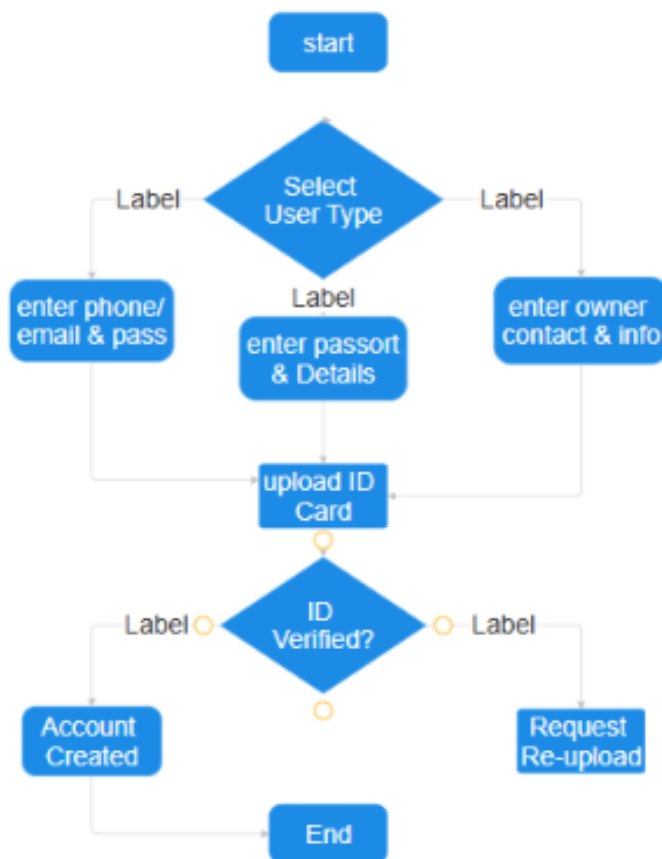
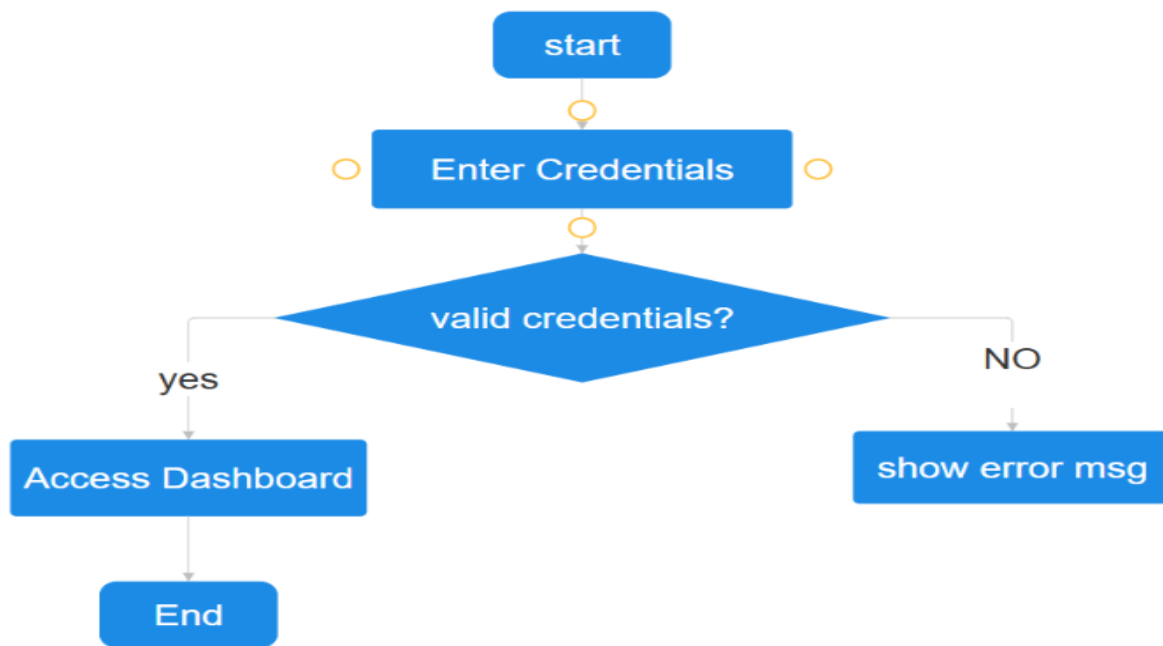


2. FIND ROOMMATE FLOW

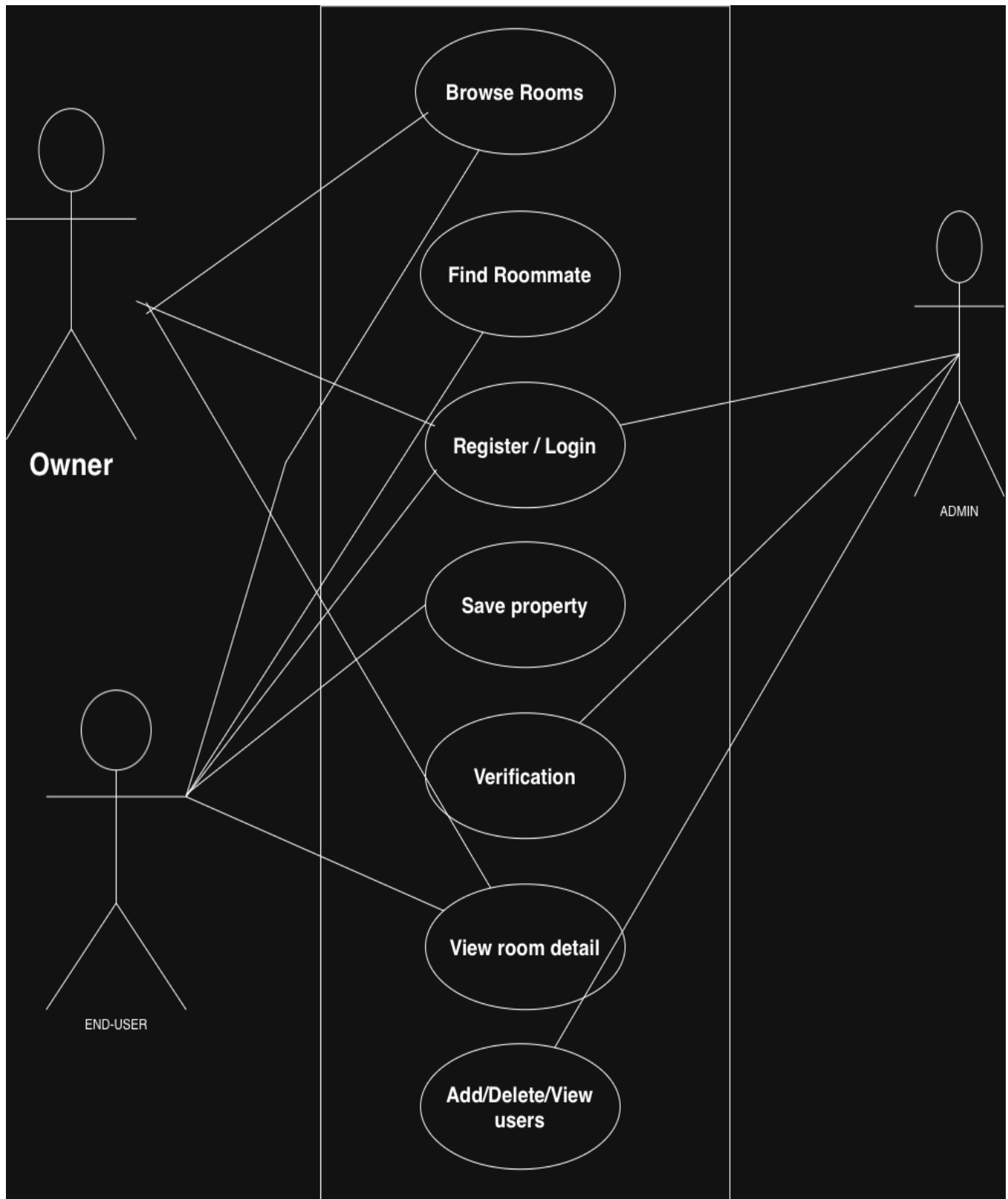


3. LISTING FLOW (POST A LISTING)





Use case Diagram



How we can make income

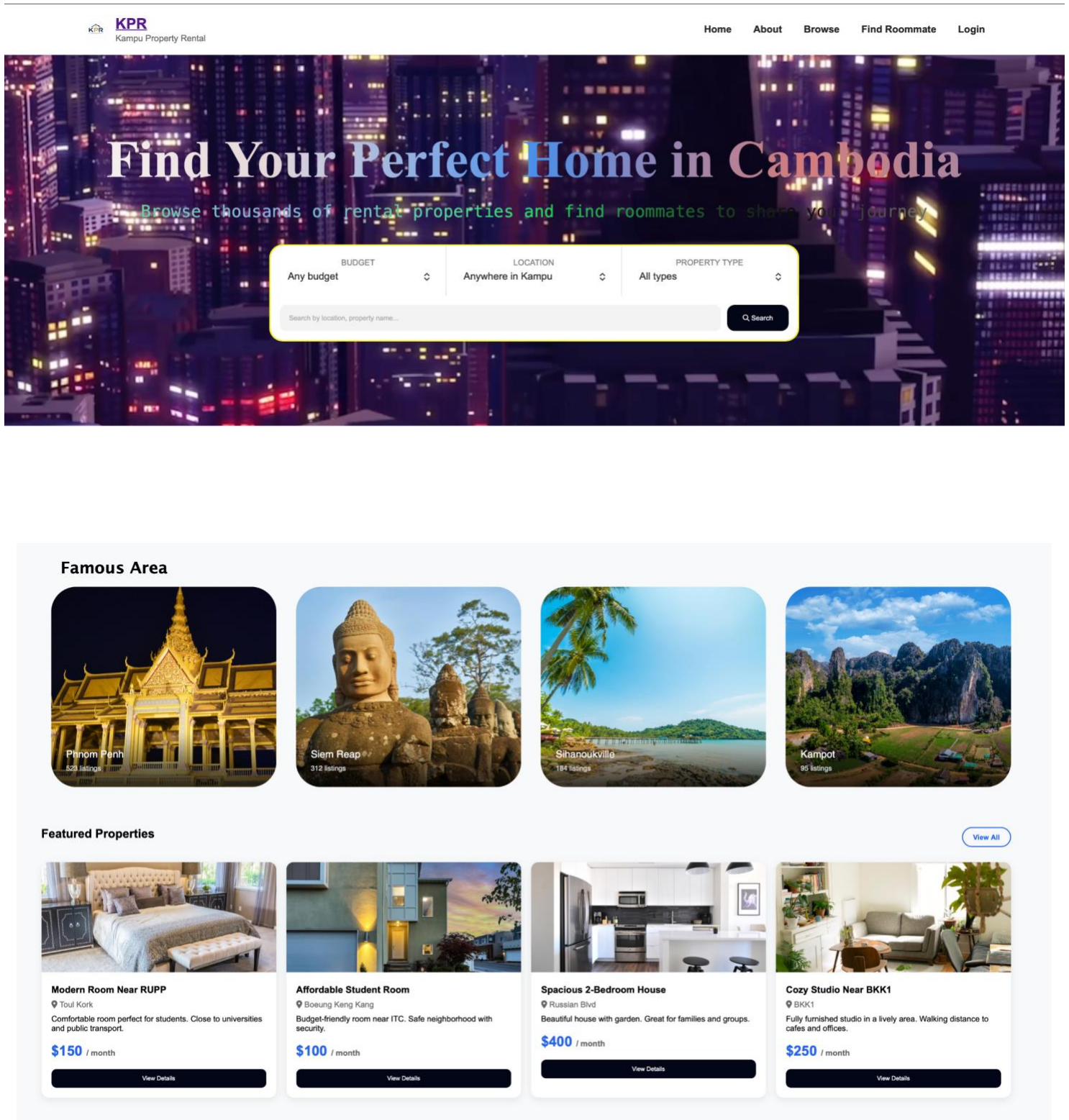
The platform generates revenue through multiple streams:

- **Advertisements from Lessors:** Lessors pay to promote their room listings, and their contents will be displayed on top of the search.
- **Free of charge:** users can post up to 3 listings for free to attract engagement.
- **Paid Posts:** by paying the subscription fees, the user is allowed to post up to 20 posts in a month and reach a wider audience. The fee is 5\$ per month.
- **Membership Plan:** A premium plan costing \$30 per year, allowing unlimited postings.

Marketing Strategy

- **Posters at Universities:** Posters are placed at universities to target students, one of the main user groups.
- **Facebook Groups:** Posting in Facebook groups helps reach a large audience quickly, as many users actively search for rental options there.
- **Partnerships:** Collaborating with real estate agents helps increase the number of listings and enhances credibility.
- **Content Creators:** Working with influencers or content creators helps promote the platform and build trust among new users.

3. System Design



Most Liked Rooms

View All



Sunny Room in Riverside Condo
📍 BKKT 34 Likes
Bright, quiet room with a great view. Includes gym and pool access.
\$220 / month
👤 Sokha



Charming Room with City View
📍 Toul Kork 97 Likes
Newly renovated, close to markets and public transport.
\$180 / month
👤 Dara



Fully Furnished Studio Apartment
📍 Chamkar Mon 84 Likes
Modern studio with kitchenette. Perfect for young professionals.
\$300 / month
👤 Rachana

1000+
Properties Listed

500+
Happy Renters

200+
Verified Landlords

150+
Roommate Matches



KPR
Kampu Property Rental

Helping you find the right room, house, or roommate across Cambodia.



Quick Links

Browse Properties
Find Roommate
Most Liked Rooms
Login

Support

Help Center
FAQ
Terms of Service
Privacy Policy

Contact

📍 Phnom Penh, Cambodia
✉️ ThyriSupports@kpr.com
☎️ +855 12 345 678

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Browse Properties

Explore rooms, houses, villas and condos across Cambodia

BUDGET
Any budget

LOCATION
Anywhere in Kampu

PROPERTY TYPE
All types

Search by location, property name...

Search

Showing 15 of 1000+ properties

Sort by: Newest



Modern Room Near RUPP
📍 Toul Kork
Comfortable room perfect for students. Close to universities and public transport.
\$150 / month
[View Details](#)



Affordable Student Room
📍 Boeung Kang Kang
Budget-friendly room near ITC. Safe neighborhood with security.
\$100 / month
[View Details](#)



Charming Room with City View
📍 Toul Kork
Newly renovated, close to markets and public transport.
\$180 / month
[View Details](#)



Fully Furnished Studio Apartment
📍 Chamkar Mon
Modern studio with kitchenette. Perfect for young professionals.
\$300 / month



Cozy Room Near Central Market
📍 Daun Penh
Walking distance to Central Market with easy access to tuk-tuks.
\$120 / month



Spacious House with Garden
📍 Sen Sok
Quiet residential area, great for families or roommates sharing.
\$450 / month

Browse Properties

Explore rooms, houses, villas and condos across Cambodia

BUDGET Any budget	LOCATION Anywhere in Kampu	PROPERTY TYPE All types
Search by location, property name...		
		Search

Showing 15 of 1000+ properties

Sort by: Newest



Modern Room Near RUPP

Toul Kork

Comfortable room perfect for students. Close to universities and public transport.

\$150 / month

View Details



Affordable Student Room

Boeung Kang Kang

Budget-friendly room near ITC. Safe neighborhood with security.

\$100 / month

View Details



Charming Room with City View

Toul Kork

Newly renovated, close to markets and public transport.

\$180 / month

View Details



Fully Furnished Studio Apartment

Chamkar Mon

Modern studio with kitchenette. Perfect for young professionals.

\$220



Cozy Room Near Central Market

Daun Penh

Walking distance to Central Market with easy access to tuk-tuks.

\$120



Spacious House with Garden

Sen Sok

Quiet residential area, great for families or roommates sharing.

\$450

Cambodia's largest roommate finder

Find a room, share your space, and connect with people across Cambodia.

HAVE A ROOM?

Need a roommate?

List your available room and find the right person.

List Your Room



LOOKING FOR A ROOM?

Create your profile

Tell us what you are looking for and find your next home.

Create Your Profile



EXPLORE

Latest Rooms & Roommates

Discover available rooms and connect with people in Cambodia.

6 Listings

All Rooms Roommates



Private Room Near BKK1

Apartment - Available Now

A comfortable private room in a modern apartment near BKK1. Close to restaurants, cafes, and markets.

BKK1, Phnom Penh



Bright Room Near Toul Kork

House - Available Now

Bright and spacious room in a quiet house in Toul Kork, close to universities.

Toul Kork, Phnom Penh



Looking for a Roommate in Siem Reap

Private Room - Looking Now

I'm looking for a friendly roommate to share a comfortable home in Siem Reap.

Siem Reap



DM ?

roommate?

able roomand find the rig

om →

Rooms & R

rooms and connect

Roommates

90/mo

m Near BKK1

able Now

rate room in a modern apartment

y to restaurants, cafes, and markets.

Bright room near Toul Kork

House • Available Now

Bright and spacious room in a quiet house in Toul Kork

close to universities.

Looking for a Roommate in

Reap

Private Room • Looking Now

I'm looking for a friendly roommate to share a

comfortable home in Siem Reap.

 **List Your Room** ×

Find someone to share your space with.

Listing Title

Example: Private room near BKK1

Location

Example: BKK1, Phnom Penh

Monthly Rent

Example: 300

Property Type

Select type

Available Date

06/08/2026

Description

Tell people about the room, location, utilities, and house rules...

Your Name

Your name

Publish Your Room

Find a room, share your space, and connect with people across Cambodia.

LOOKING FOR A ROOM?

ed a roommate?

our available roomand find the rig

Your Room →

LOOKING FOR A ROOM?

Create your profile

at your next

Rooms

Roommates

300/mo

Room Near BKK1

Available Now

able private room in a modern apartment

K1. Close to restaurants, cafes, and markets.

Phnom Penh

180/mo

Bright Room Near Toul Kork

House • Available Now

Bright and spacious room in a quiet house in Toul Kork

close to universities.

Toul Kork, Phnom Penh

180/mo

Looking for a Roommate in Siem Reap

Private Room • Looking Now

I'm looking for a friendly roommate to share a

comfortable home in Siem Reap.

Siem Reap

 **Create Your Profile** ×

Tell people what kind of place you are looking for.

Full Name

Your full name

Age

Age

Gender

Select

Looking in

Example: Phnom Penh

Monthly Budget

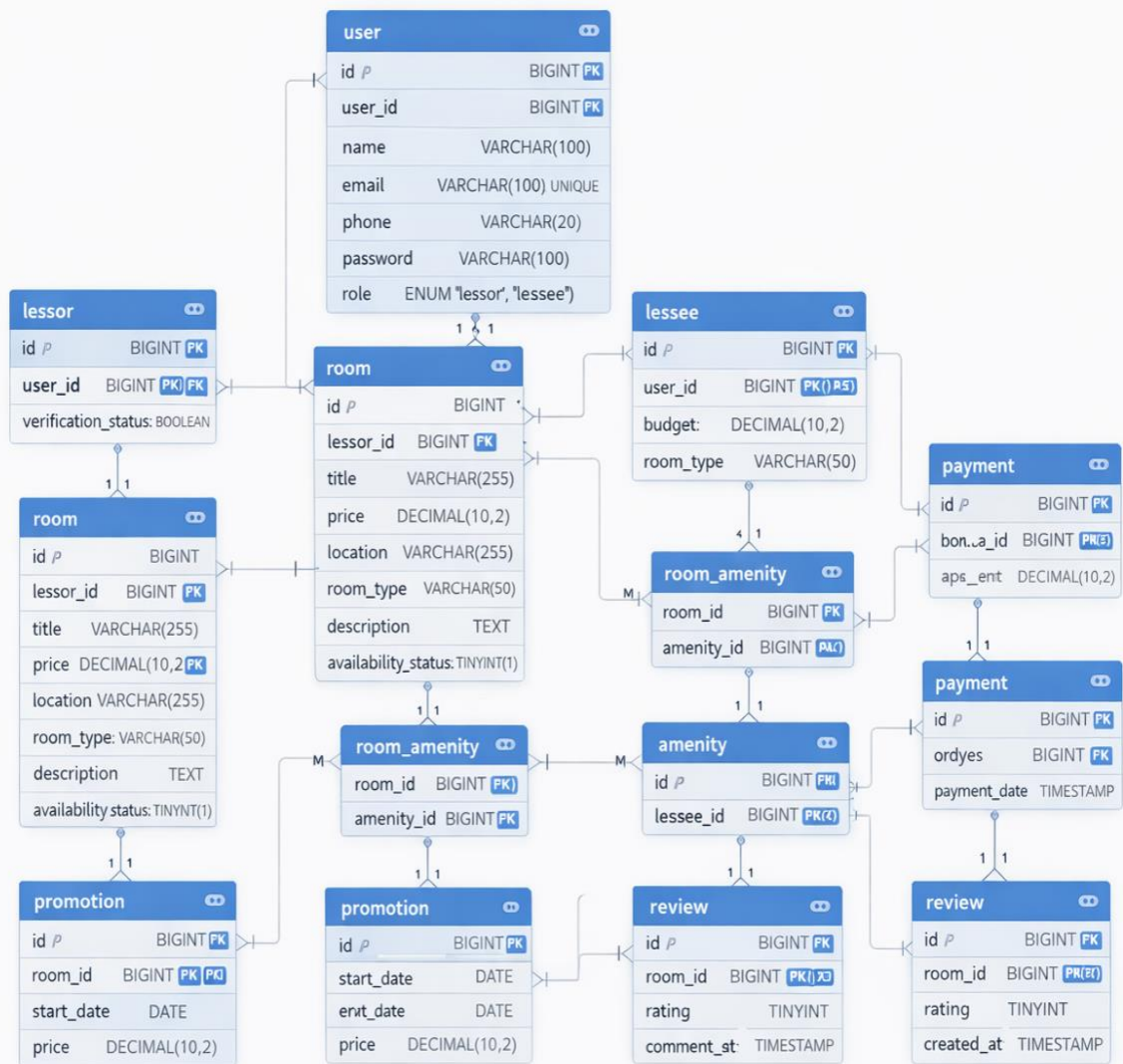
Example: 300

About You

Tell potential roommates about yourself...

Create My Profile

ER Diagram



Kampu Property Rental System – ER Diagram

Methodology

As for the development of our project, we are using the **Agile** model. The development process will start with the identification of the needs lasting for 2 months, during that phase, we focus on defining the project scopes, collecting the requirements, and designing the system models such as ER diagrams, Flowchart, and UI Wireframes. This phase is vital as it ensures a clear roadmap for the development. After we finish the project research, we will be executing the Agile, which will probably last for 12 to 15 months. The development journey will be divided into small segments and increase over time. Finally, the final 1-month period phase is dedicated to the software testing to test on the performance optimization, security audits, and final deployment.

Development tools

- Front-end:
 - Languages: HTML, CSS, JavaScript
 - Framework: Bootstrap css, React js
 - Code Editor: Visual Studio Code (VScode)
 - Version Control: Git, GitHub
- Back-end:
 - Database: MySQL
 - Server: [Node.js](#) with [Express.js](#)
- Hosting:
 - Cloud: DigitalOcean

Project Timeline

Time Line	Activities
Week 1 and 2	Identify the topic and discussion with the group members
Week 3 to 5	Collecting required information to define the topic
Week 6 to 8	Designing the UI, Prototypes, ER and flowcharts
For next 2 months	Implementing the front-end part and with regular meeting testing
For the next 8 months	Implement all of the essential functions
For the next 2 months	Implementing the security checking, and performance optimization
For the last 1 month	Finalized the system testing, and fix all the bugs

Cost Analysis

The cost analysis was made by using COCOMO model, with the expectation that there will be around **22 thousand lines of code** and the project will be Semi-detached project type due to medium complexity with mixed experience of developers to reduce time taken in creating UI in developing the application. Below will be the calculation in details:

Category	a_b	b_b	c_b	d_b
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Since our project will be semi-detached project type, we will follow the default provided value.

Calculate the Effort(E):

$$E = a * (KLOC)^b$$

$$\delta E = 3.0 * (22)^{1.12} = \mathbf{96 \text{ person-months}}$$

· Development Time (T):

$$T = c * (E)^d$$

$$\delta T = 2.5 * (96)^{0.35} = \mathbf{12 \text{ months}}$$

· Require team:

According to this calculation model, the team members need to develop this project will be approximately 8 persons, but because of the budget concern, this project will have a development team of 3 people, and the duration will be increased to 15 months to ensure a successful project. The role will describe as below:

Position	Number	Monthly Salary	Duration (In total)
Full-Stack Developer	1	1,200 USD	15 Months
Frontend Developer	1	800 USD	10 Months
QA	1	700 USD	3 Months

$$\Rightarrow \text{Total Personnel cost} \Rightarrow 28,100 \$$$

- Infrastructure cost:

Since the server will be host on DigitalOcean, therefore, the cost will be as describe below:

- Monthly: 6\$ /month
 $\Rightarrow \text{Total} = 6 * 15 = \mathbf{90\$}$

In conclusion, the total cost of this project will show in the table below:

Category	Cost (USD)	Cost (KHR)
Personnel	28,100 \$	116,805,000 KHR
Software Tools	0 \$	0 KHR
Cloud Hosting	90 \$	364,500 KHR
Total	28,190 \$	117,169,500 KHR